

Theme Discovery from Scratch

Motivation, taxonomy, and lifecycle

May 27, 2026

Thematic trading

Why thematic trading?

Traditional investing organises companies along **time-invariant** axes: geography, sector, market cap, style (value/growth).

- Easy to benchmark, easy to index.
- Slow to capture *structural shifts* that cut across sectors.

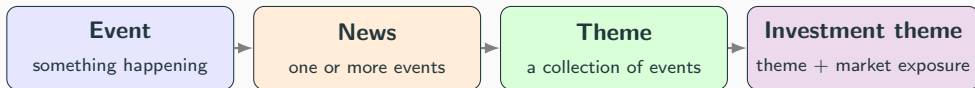
Thematic investing organises companies along a **narrative**: a structural change driving demand across multiple sectors at once.

- AI infrastructure $\Rightarrow$ chip designers + utilities + cooling + real estate
- Energy transition $\Rightarrow$ miners + grid operators + auto manufacturers
- GLP-1 obesity drugs $\Rightarrow$ pharma + food & beverage + medical devices

*The bet: catching the narrative **before** it is priced into every constituent.*

Taxonomy for theme detection

From news to theme — granularity ladder



- **News** is the raw input — articles bundle multiple events that contains one or more events.
- **Events** are single, datable happenings (a product launch, a policy announcement, an earnings beat).
- **Themes** are recurring patterns over events — a narrative binding them together.
- **Investment themes** add the second leg: a tradable basket whose value moves with the theme.

Sectors — the time-invariant baseline

Sectors classify *companies* along stable, slow-moving axes. Several competing taxonomies exist:

Standard	Maintained by	Granularity
GICS	MSCI / S&P	11 sectors → 25 industry groups → 74 industries
ICB	FTSE Russell	11 industries → 20 supersectors → 45 sectors
RBICS	FactSet	~1,500 leaf-level sub-industries (revenue-based)

Key property: a company's sector code rarely changes: **Time Invariant**.

What sectors cannot capture: cross-sector narratives, transient shifts, and companies whose thematic exposure is much larger than their sector classification suggests.

What is a theme? Two defining properties

A **theme** is a coherent narrative that ties together a stream of events. Two properties:

1. Transient (has a lifecycle)

Sectors are roughly permanent.

Themes emerge, accelerate, mature, and fade.

- *Cannabis* (2018–2021), now dormant
- *Metaverse* (2021–2022), faded fast
- *GLP-1* (2023–), still accelerating

2. Cross-sector

A theme groups companies that a sector taxonomy would scatter.

- *AI infrastructure*: semis (NVDA), utilities (VST), real estate (DLR), industrials (ETN)
- *Energy transition*: miners, grid operators, auto, chemicals

Operational definition for this project: a theme is a cluster of *events* that recur together over a sustained window and are mentioned in news with rising frequency.

Theme vs. investment theme

Theme

A narrative observable in the world.

- *The data exists*: news mentions, event clusters, co-occurrence patterns.
- Detectable from a news corpus alone.
- No requirement for any tradable instrument to exist.

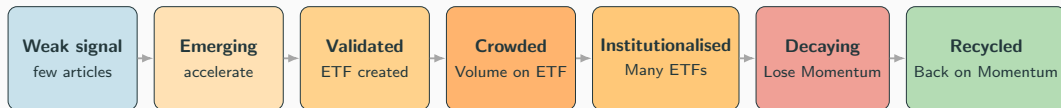
Investment theme

Theme + a defined market exposure.

- A *basket* (or ETF, or screen) whose value moves with the theme.
- Requires mapping the theme to companies (and weights).
- Trade-off: tighter basket $\Rightarrow$ more upside if right, but more risky.

Many themes never become investment themes (no clean basket exists, or the public-market exposure is too small). *Theme detection comes first; basket construction is a separate problem.*

The lifecycle of a theme



The tension: the earlier you detect, the more upside — but also the more noise. Most “weak signals” don’t become themes.

Where we want to operate: emerging → validated.

Clean Energy — full lifecycle (2018–2024)

Cross-sector by construction.

One narrative, beneficiaries in five different GICS sectors:

- **Utilities** — NextEra (NEE), Iberdrola (IBDRY)
- **Industrials** — Vestas (VWS), Enphase (ENPH)
- **Materials** — Albemarle (ALB), SQM (lithium)
- **Auto / consumer disc.** — Tesla (TSLA), BYD
- **Tech** — solar inverters, grid software

Lifecycle stage by stage.

- *Weak signal* 2018: niche ESG flows, falling solar cost
- *Emerging* 2019: Paris Agreement bites, EU Green Deal
- *Validated* 2020: Biden wins on climate platform
- *Crowded* early 2021: ICLN, TAN go vertical; HF flows
- *Institutionalised* 2021: every diversified book has ICLN
- *Decaying* 2022: rate hikes, supply-chain hits, peak euphoria fades

What you missed if you waited for the ETF stage.

ICLN (iShares clean-energy ETF): ~\$10 early 2020 → ~\$33 Jan 2021 (~3.3×).

TSLA: ~\$30 split-adj. early 2020 → ~\$400+ at peak.

ENPH: ~\$25 early 2020 → ~\$330 peak.

Most of the return happened in the *emerging* → *crowded* window (2019–early 2021).

The recycled wave (2023–24). Old constituents that decayed (TSLA, ENPH) didn't recover. But *nuclear utilities* (VST, CEG) — barely in the original basket — carried the theme's energy-transition logic forward under the AI label.

AI — winning theme (2022–2024)

Cross-sector beneficiaries.

Obvious in semis — but the real edge came from the second-order plays:

- **Semis / hardware** — NVDA, AMD, TSM, AVGO
- **Software** — MSFT, GOOG, ORCL
- **Utilities / power** — VST, CEG, TLN (AI data-centre power demand)
- **Industrials** — VRT, ETN (cooling, electrical infra)
- **Real estate** — DLR, EQIX (data centres)
- **Materials** — copper miners (FCX) for grid build-out

Lifecycle stage by stage.

- *Weak signal* 2017–2019: Transformer paper, GPT-2; niche
- *Emerging* 2020–early 2022: GPT-3, scaling laws
- *Validated* Nov 2022: ChatGPT moment — narrative crystallises
- *Crowded* 2023: NVDA +200% over the HF long the

What catching the narrative early was worth.

NVDA: ~\$15 Oct 2022 → ~\$135 mid-2024 (~9×).

VST (utility, AI-power play): ~\$25 early 2023 → ~\$170 peak (~7×). Sector ETF XLU did ~+10% over the same window.

The *utilities* leg is the key insight: VST is in no AI-themed ETF at the start of 2023, invisible to a sector view. A cross-sector thematic lens was required to surface it.

Metaverse — counterexample (2021–2022)

Looked like a textbook cross-sector theme.

- **Tech** — Meta (FB→META), MSFT (HoloLens)
- **Gaming** — Roblox (RBLX), Unity (U)
- **Hardware** — Apple (Vision Pro), Sony
- **Consumer** — Nike (NFT push), luxury brands
- **Semis** — NVDA (Omniverse pitch)

Lifecycle that short-circuited.

- *Weak signal* 2014–2020: VR niche, Oculus acquired by Facebook
- *Emerging* Oct 2021: Facebook renames to Meta; Roundhill META ETF launches
- *Crowded* late 2021: peak hype, every CEO mentions it
- **Validated stage skipped**
- *Decaying* 2022: Reality Labs losses balloon
- *Recycled?* arguably as “spatial computing” / Apple Vision Pro — but mostly dormant

What “buying the theme” cost.

META (the stock): peak ~\$380 Sep 2021 → trough ~\$90 Nov 2022 (−75%).

RBLX: ~\$140 peak → ~\$25 trough.

META (the Roundhill ETF): launched near peak hype, never recovered.

Cross-sector *and* news velocity were both present. **What failed was validation**: the underlying demand never materialised. A pure news-acceleration signal would have flagged this — and been wrong.

Quantum Computing & diamonds — today's weak signal

The narrative.

Leading quantum-hardware architectures (NV-centre qubits, photonic quantum sensors) need **ultra-pure synthetic diamond substrates**. CVD-diamond producers become indirect quantum plays.

Potential cross-sector beneficiaries.

- **Tech** — IBM, IonQ, RGTI (the names everyone watches)
- **Materials / mining** — synthetic-diamond producers (mostly private; a few small caps)
- **Semis** — quantum-control electronics
- **Defense** — quantum sensing (navigation, ASW)

Lifecycle position today

Broad “Quantum” theme: *emerging* (1–2 ETFs launched, niche coverage).

“Diamond supply chain for quantum” sub-theme: *weak signal*.

A sector view would never surface mining companies as quantum plays.

The honest question. Both Clean Energy and Metaverse looked exactly like this at the weak-signal stage. Where do you want the precision/recall threshold of the system?

Tracking themes over time

Static models find clusters *after the fact*. **Dynamic** models show that topic vocabulary moves. An **online / candidate-theme** system answers the harder question: is today's cluster the *continuation* of yesterday's, a *split*, a *merger*, or genuinely *new*?

Identity vs. vocabulary drift

Topic *identity* stays fixed; *vocabulary* moves.
D-ETM-style models share embedding structure so semantically related words drift together — rather than every rare word having to earn its own history.

Online tracking (BERTrend-style)

Refit/extend the global map only with documents *known as of the decision date*. Keep an **outlier buffer** for proto-themes that have not yet stabilised into a cluster.

Four transitions to detect

Continuation	same theme, drifted vocabulary
Split	one theme branches into several
Merge	several threads converge
Birth	genuinely new candidate theme

False positives to guard against

Duplicate documents, a new source added to the corpus, ticker / OCR errors, seasonal language, coordinated manipulation, model retraining itself.

Better embeddings for the sector-aware pipeline

Today (0.3-themes.ipynb). FinLang embeds news headlines and 173 ICB subsector definitions → averaged into **45 sector centroids** → BERTopic on headline embeddings → cosine vs centroids → *aligned* ($\max_{\text{cos}} \geq \tau_{\text{cov}} = 0.30$) / *novel*. Last run: 22 aligned, 18 novel. *Encoder choice is the single biggest lever — a domain-tuned swap typically beats any pooling or preprocessing tweak.*

Encoder candidates (biggest lever)

- **FinLang** — current baseline (finance-tuned).
- **FinBERT-MPNet** — head-to-head with FinLang on the same 45-sector cosine task.
- **LoRA on bge-large** — strong general encoder lightly adapted to financial text; cheaper to maintain than full domain pretraining.

Pooling & preprocessing

- **Sum-pool / concat last 4 hidden layers** of the SBERT encoder, instead of default last-layer mean pooling — cited +10–20% *relative* on topic diversity & coherence on news corpora.
- **Stop-word removal at embedding time**, not only at the c-TF-IDF stage — small alone, compounds with the rest.

Evaluation on the current pipeline. Topic coherence (NPMI, C_v) and diversity on BERTopic output; sector separation on the 45-centroid axis — median `max_cos` for *aligned* vs *novel*, and the share of themes crossing τ_{cov} .